IE4-DS LAB SESSION 1 – REPORT

Synchronous Serial Communication with an Audio Codec

Team (group nummer + team letter as chosen in Moodle, e.g. 02 A)	**Team members**	
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	2	
	3	

Please fill out the document electronically – including the yellow fields above – and upload it to Moodle once per team at the latest 30 min after the end of the lab session (no matter who uploads it, it will automatically be assigned to all members). Please note that the complete assignment can be found in another document.

Lab Task 1: Test of the system with a sinusoidal signal

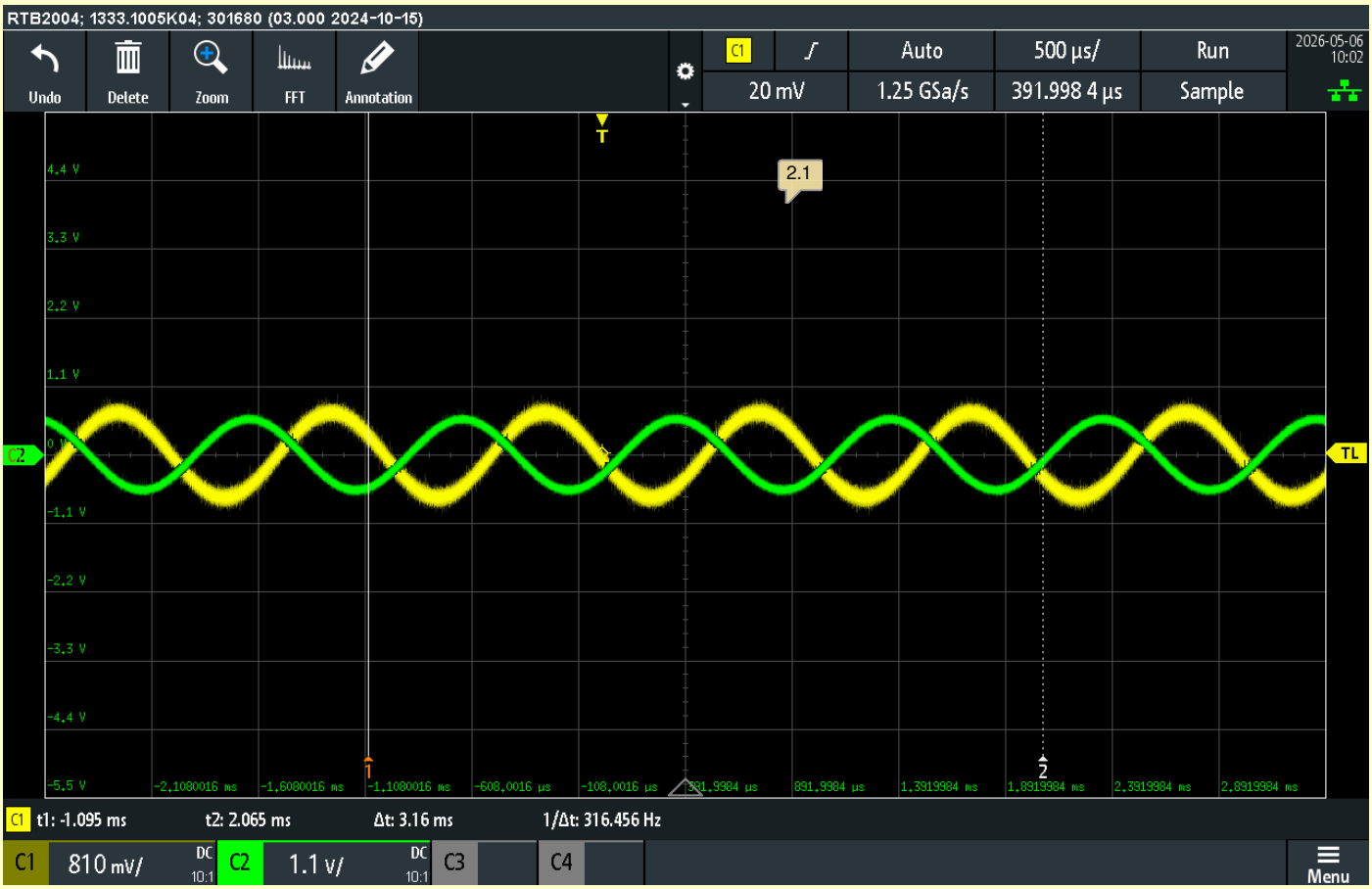
With this test you can see if you have implemented everything correctly. The input signal at

...

output still does not meet the expectation, continue first with Lab Task 2 for troubleshooting and return later.

- Apply a sinusoidal voltage of e.g. 1 V and 1 kHz to the LINE IN input. Check the amplitude and frequency with the oscilloscope before connecting the function generator. Display both, the input and output signal, on the oscilloscope.
- If the output signal was as expected, slowly turn up the frequency of the sinusoidal

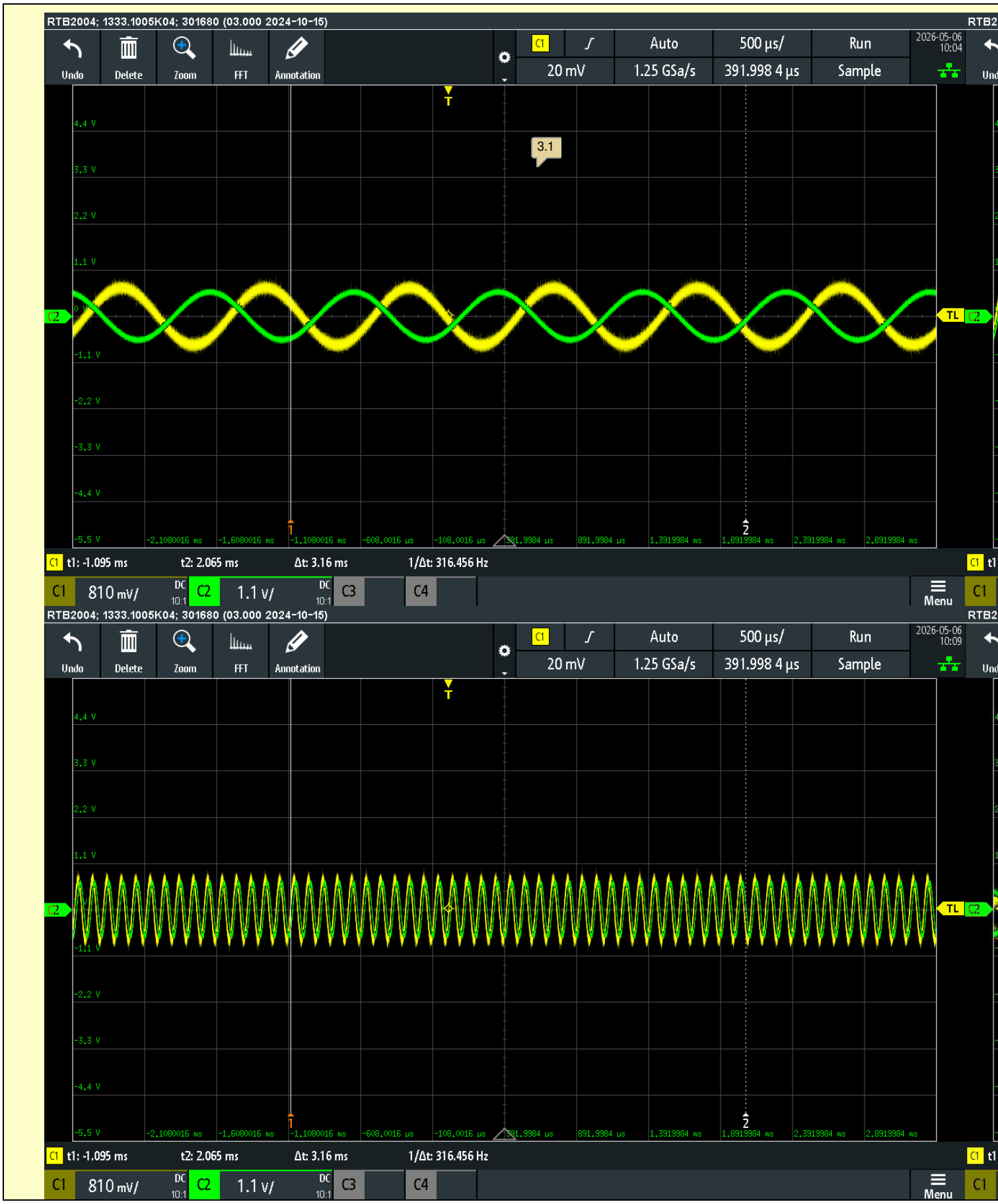
- **Oscilloscope screenshot:**



Display both, the input and output signal, on the oscilloscope.

- If the output signal was as expected, slowly turn up the frequency of the sinusoidal signal to 30 kHz. Describe what you observe and name some characteristic frequency and amplitude values.

- **Oscilloscope screenshots showing the behavior for higher frequencies and verbal description**


RTB2004; 1333.1005K04; 301680 (03.000 2024-10-15)
Undo
Delete
Zoom
FFT
Annotation
C1
Auto
500 µs/
Run
20 mV
1.25 GSa/s
391.998 4 µs
Sample
2026-05-06
10:04
4.4 V
3.3 V
2.2 V
1.1 V
-1.1 V
-2.2 V
-3.3 V
-4.4 V
-5.5 V
3.1
C2
TL
t1: -1.095 ms
t2: 2.065 ms
Δt: 3.16 ms
1/Δt: 316.456 Hz
C1 810 mV/
C2 1.1 V/
C3
C4
Menu
RTB2004; 1333.1005K04; 301680 (03.000 2024-10-15)
2026-05-06
10:09


RTB2004; 1333.1005K04; 301680 (03.000 2024-10-15)
Undo
Delete
Zoom
FFT
Annotation
C1
Auto
10 µs/
Run
20 mV
1.25 GSa/s
391.998 4 µs
Sample
2026-05-06
10:14
4.39 V
3.29 V
2.19 V
1.09 V
-1.11 V
-2.21 V
-3.31 V
-4.41 V
-5.51 V
341.9984 µs
351.9984 µs
361.9984 µs
371.9984 µs
381.9984 µs
391.9984 µs
401.9984 µs
411.9984 µs
421.9984 µs
431.9984 µs
441.9984 µs
C2
TL
C1 t1: 362.3 µs
t2: 425.5 µs
Δt: 63.2 µs
1/Δt: 15.8228 kHz
C1 810 mV/
DC 10:1
C2 1.1 V/
C3
C4
Menu
RTB2004; 1333.1005K04; 301680 (03.000 2024-10-15)
2026-05-06
10:16

Starting at 1 kHz, the LINE OUT signal (Ch2, green) appears as a time-shifted version of the LINE IN s
that the system passes audio frequencies in this range without significant attenuation.

Above approximately 20 kHz a sharp amplitude roll-off is observed on LINE OUT. At 25 kHz the output

This behaviour is caused by the **anti-aliasing low-pass filter** integrated in the PCM3006 audio codec
ing during the ADC conversion process. The characteristic cutoff frequency of this filter is therefore app

Lab Task 2: Analysis of serial bit streams

In this task, we look at the serial, digital data that is exchanged between the FPGA and the audio codec. If Lab Task 1 did not bring the desired results, here is the possibility to make an

- Apply a sinusoidal voltage of e.g. 1 V and 20 kHz to the LINE IN input. Take a complete data frame of DIN and DOUT together with the BCK and LRC signal. One data frame corresponds to one complete period of the LRC signal with one sample each for left and right channel.

- **Oscilloscope screenshot**





- Determine the 16 bit sampling values for left and right channel and express them as decimal numbers. Are the values of DIN and DOUT the same? Explain.

- **Abtastwerte links / rechts:**

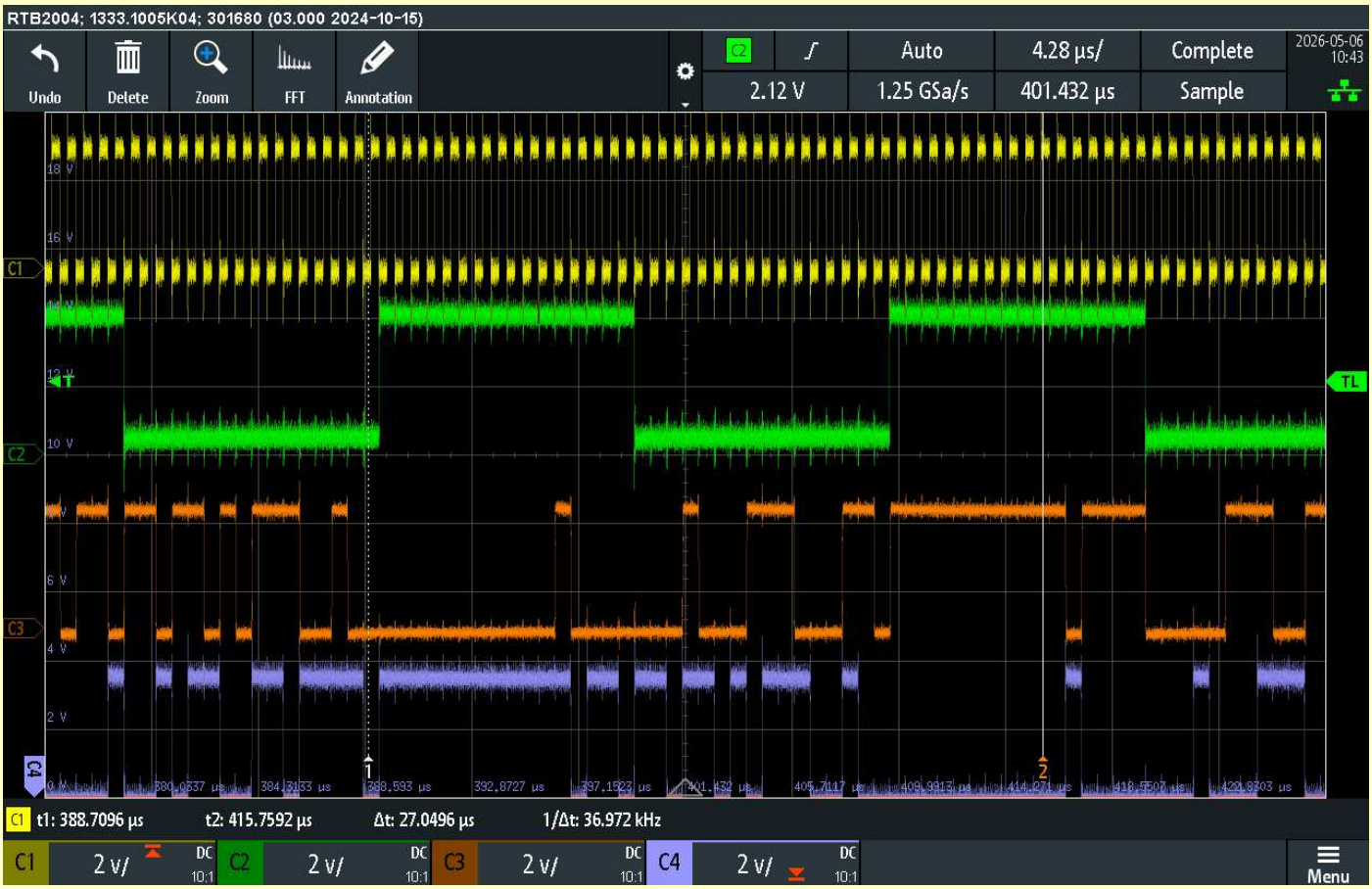
  6.2

  No. Within the same LRC frame, DIN = −32768 and DOUT = 0. They differ because the system introduces exactly one LRC period of pipeline delay (≈ 20.27 µs). DOUT in frame N carries the data that was received on DIN during frame N−1. The AUDIO_PROCESSING_DUMMY module passes data unchanged but only after a full deserialisation and re-serialisation cycle through SER_TO_PAR_LR and PAR_TO_SER_LR.

  6.3

- Record two consecutive data frames. What can you now state about DIN and DOUT?
- Document the results with oscilloscope images (screenshots, but *not* cell phone photos).

- **Oscilloscope screenshot:**



- **Explanation:**

  The oscilloscope capture at 4.28 µs/div shows approximately two and a half consecutive LRC frames. Two complete LRC periods are clearly visible in Ch2 (LRC, green), each spanning approximately 20.3 µs. Comparing Ch3 (DIN, orange) between frame N and frame N+1, the bit pattern changes clearly — this is expected since the 20 kHz sine wave advances by approximately 0.4 of a full period between consecutive samples at 48 kHz, resulting in a significantly different sample value each frame.

  Comparing Ch4 (DOUT, blue) in frame N+1 with Ch3 (DIN, orange) in frame N, the bit patterns show strong similarity, confirming that DOUT echoes DIN with a delay of exactly one LRC period. This one-frame latency is an inherent property of the serial interface pipeline: data received on DIN during frame N is deserialised, passed through AUDIO_PROCESSING_DUMMY, re-serialised, and output on DOUT during frame N+1.

7.1

Lab Task 3: Bit Timing

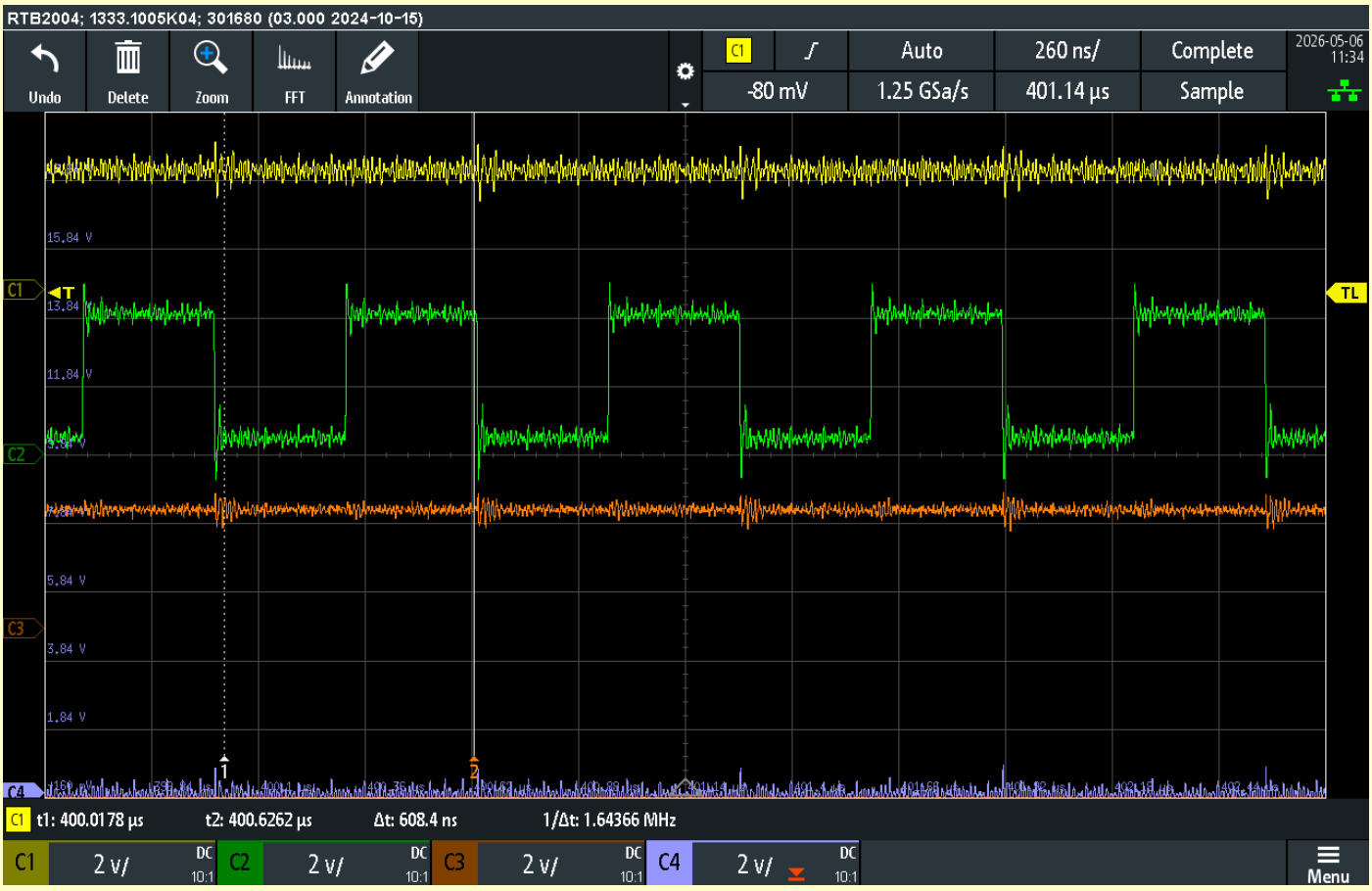
In this part you are to check whether the specifications of the data sheet for the timing behavior of the signals are met. The signal names according to the left column in table1.3 are used in the following. Measure the following times and compare them with the data

- BCK pulse cycle time $t_{(BCY)}$ (measure at BCK)
- DOUT delay time to BCK falling edge $t_{(BDO)}$ (measur at BCK and DIN)
- DIN setup time $t_{(DIS)}$ (measure at DOUT and BCK)
- DIN hold time $t_{(DIH)}$, (measure at DOUT and BCK)

Only measure two of the four times. You can choose freely, but indicate clearly which ones you have checked for complying with the specifications.

- **Oscilloscope screenshots of both measurements**

  BCK pulse cycle time t(BCY):



  DOUT delay time to BCK falling edge t(BDO) (measur at BCK and DIN)



- **Are the two measured times in accordance with the datasheet specification? Explain.**

  Looking at the waveform, cursor 1 is on a BCK rising edge and cursor 2 is on the next BCK rising edge.

  Therefore:

  t(BCY) = 608.4 ns — this is the full BCK cycle time.

  From the same waveform I can estimate:

  - **BCK HIGH time (from rising to falling) ≈ half of 608.4 ns = ~304 ns = t(BCH)**
  - **BCK LOW time (from falling to rising) ≈ half of 608.4 ns = ~304 ns = t(BCL)**

t(BDO) = 25.2 ns — datasheet max 40 ns . Pass. 10.1

Lab Task 4: Delay of the overall system

First measure the time delay between the analog LINE IN input and the LINE OUT output. Then you are to estimate which part of the delay is due to the audio codec and which part is due to the FPGA.

- What is the risk if you measure the delay with a sinusoidal signal?

- **Explain.**

  A sine wave repeats every period, so a delay of Δt is indistinguishable from Δt + n×T. The narrow rectangular pulse has a unique rising edge per period, making the delay unambiguous

- Therefore, perform the measurement with a rectangular 1 kHz pulse with a duty cycle of 10% and ±1 V amplitude. How do you measure the delay you are looking for?
- Explain the shape of the output signal. Is it still rectangular?
- Which part of the delay is due to the audio codec, which due to the FPGA? Use the results of Lab Task 2 for your considerations, too.

- **Oscilloscope screenshots of the measurement**



No. The PCM3006 low-pass filters (cutoff ~20–24 kHz) remove the high-frequency harmonics of the rectangular pulse, producing a smoothed, rounded output waveform.

- **Analyse the delay times and explain**

  The codec (ADC + DAC filters) accounts for ~577 µs (96.6% of total). The FPGA contributes exactly one LRC period (~20.5 µs, confirmed in Lab Task 2), only 3.4% of the total delay. The dominant delay source is the PCM3006 delta-sigma filter chain, not the FPGA.